INCH-POUND

A-A-55172
04 February 1993

COMMERCIAL ITEM DESCRIPTION

LAST, FOOTWEAR, FIELD, WOMEN'S

The General Services Administration has authorized the use of this Commercial Item Description in preference to MIL-L-1741.

Abstract. This document covers the requirements for plastic footwear lasts. The lasts are intended for use as a base over which a single design and quality of women's field footwear for military personnel of the Department of Defense may be manufactured.

Salient Characteristics. The construction of the lasts shall conform to Government furnished models, patterns and templates and shall be as specified herein.

Material.

Polyethylene Blocks. The lasts shall be made from polyethylene blocks. Polyethylene blocks shall be made from solid high density polyethylene without fillers, conforming to types III or IV (except that the density of type IV shall not exceed 0.965), category 4 or 5 of ASTM D 1248 (melt index 1.0 or less when tested as specified in ASTM D 1248). The color shall be the standard color used by the supplier. The blocks shall be made from virgin material of the same composition produced by regrinding clean unburned scrap produced in the fabrication of last blocks, finished lasts and imperfect lasts. Scrap containing dirt or other foreign material shall not be used.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FSSD, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8335

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Metal Hinges. The metal hinges shall be a commercial type, made of steel, and zinc chromated to resist moisture.

Hinge Pins. The hinge pins shall be made of steel wire and shall be heat treated to a hardness value of 45 to 50 Rockwell C scale when tested as specified in test method 243 of FED-STD-151. The pins shall be not less than 15/64 inch or more than 19/64 inch in diameter. The pins shall be countersunk 1/4 (± 1/8) inch under the surface of the sides of the lasts. The pins shall be zinc chromated to resist moisture and shall be mechanically peened.

Thimble. The thimble shall be manufactured from steel sheet or tubing stock and shall be heat treated to a hardness value of 55 to 88 Rockwell B scale when tested as specified in test method 243 of FED-STD-151. The thimble shall be of the drawn, split, or swaged tubing type. All thimbles shall be full-flanged with a closed or a partially closed bottom, and shall be the locked-in type. Dimensions for last thimbles shall correspond to the following:

Inside diameter	0.500 ± 0.046 inch
Overall length	1.625 ± 0.063 inches
Depth of hole	1.500 ± 0.063 inches
Wall thickness	0.0475 to 0.095 inch
Flange diameter	0.750 ± 0.063 inch

Heelplate. The heelplate shall be not less than 2-7/16 inches long on size 4B, shall measure proportionate to normal grade on other sizes, and shall have a 1/2 (± 1/32) inch hole in the center for insole tacking and five countersunk nail holes. The heelplate shall be made from either hot-galvanized or electro-galvanized sheet steel 0.050 (± 0.005) inch thick, and shall be treated to a hardness value of 48 to 67 Rockwell B scale, when tested as specified in test method 243 of FED-STD-151.

Plastic Plug. The plug shall be a composition plastic which shall be either of the same composition as the body of the plastic last or of a special composition meeting the requirements below. The plug shall be 1/2 (± 1/32) inch in diameter and 3/8 (± 1/16) inch long. When of special composition, the hardness value as measured by the Shore durometer shall be not less than 60 D-scale when tested as specified in test method 1082 of FED-STD-406. The plastic heel plug of either composition specified above shall be a force fit in the drilled tack hole, and shall show no evidence of splitting when the plastic heel plug in the finished last is subjected to the insertion of a 2-1/2 ounce shoe tack for at least 1/4 inch.

Heelplate Attaching Nails. The nails for attaching the heelplate on plastic lasts shall be 15 or 16 gage, 5/8 to 3/4 inch long, barbed, iron wire nails with 1/8 (± 1/32) inch diameter flat heads and shall have a ringed chisel point and shall not be cement coated.

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Ink. Ink for marking the plastic lasts shall be black, shall not blur and shall be set permanently into the plastic by heat. The finished last shall be wet in a test area including marking with acetone for 5 minutes (e.g. using soaked cotton wad). The marking shall be tested by attempting to scrape through the test area with the thumbnail. No change in marking shall occur.

Construction.

Hinge Cutting, Slotting and Assembly. The hinge V-cut and circle sawcut shall correspond to the hinge pin holes so that the last, with a correctly inserted hinge, will open and close in a satisfactory manner. All inside sharp corners on the plastic lasts shall have a slight radius made by running over the edge with a hot wire to eliminate stress risers made by cutters during the hinge cutting operation. The use of a radius ground cutter to eliminate stress risers shall be permitted.

Thimble Hole. The thimble hole shall be bored to such a size that when the thimble is inserted in the plastic last, it shall be locked into the plastic. Thimbles shall be countersunk on plastic lasts. On sizes 7-1/2 through 12 (all widths), the thimble hole shall be centered 1-1/2 ($\pm$1/16) inches from the back of the last top. On sizes below 7-1/2, the holes shall be centered 1-3/8 ($\pm$1/16) inches from the back of the last top.

Heelplating. The heelplate on the finished last shall be flush with, but not overlaying, the outside edge of the last. A narrow margin of plastic, having a maximum width of not more than 1/16 inch in the curved area beginning 3/8 inch from the breast line of the plate, is acceptable. The margin of plastic in the area 3/8 inch from the breast line on each side of the plate is not limited. The breast of the heelplate shall butt up to the plastic. The heelplates shall be attached with five of the nails specified above. The position of the heel nails shall be according to industry practice with countersunk nail holes punched not less than 3/8 nor more than 5/8 inch from the edge of the plate except that the two front nail holes shall be not less than 3/16 nor more than 3/8 inch from the breast of the heelplate. The side nails shall be located equidistant between the breast nails and the back nail. The heelplates shall be punched with a 1/2 ($\pm$1/32) inch tack hole located approximately in the center of the heelplate, measuring both from breast to the end of the heelplate and from side to side. The heelplates shall be cupped to conform to the contour of the heel seat of the last.

Tack Hole Plugging. Tack hole plugging for the lasts shall be done with composition plastic plug specified above. The plug shall be inserted so that it entirely fills the 1/2 ($\pm$1/32) inch hole which is drilled through the heelplate to a depth of not less than 3/8 inch nor more than 1/2 inch. After the plug has been inserted, it shall be filed, shaved, or ground even with the heelplate.

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Heel and Toe Making. The heel and toe shall be shaped following the heel and toe profile templates.

Sizes and Widths. The lasts shall be provided in the sizes and widths shown in Table I below.

Table I. Schedule of Sizes

Schedule of widths	AAAA	AAA	AA	A	B	C	D	E
2-1/2	-	-	-	X	X	X	X	X
3	-	-	X	X	X	X	X	X
3-1/2	-	X	X	X	X	X	X	X
4	X	X	X	X	X	X	X	X
4-1/2	X	X	X	X	X	X	X	X
5	X	X	X	X	X	X	X	X
5-1/2	X	X	X	X	X	X	X	X
6	X	X	X	X	X	X	X	X
6-1/2	X	X	X	X	X	X	X	X
7	X	X	X	X	X	X	X	X
7-1/2	X	X	X	X	X	X	X	X
8	X	X	X	X	X	X	X	X
8-1/2	X	X	X	X	X	X	X	X
9	X	X	X	X	X	X	X	X
9-1/2	X	X	X	X	X	X	X	X
10	X	X	X	X	X	X	X	X
10-1/2	X	X	X	X	X	X	X	X
11	X	X	X	X	X	X	X	X
11-1/2	X	X	X	X	X	X	X	-
12	-	X	X	X	-	-	-	-

Marking.

Size and Width Stamping. The sawcut of the heel part shall be clearly and legibly rubber-stamped with the proper size of the last, using ink specified above. The numerical size shall be stamped on the left of the hinge cut and start no more than 3/16 inch from the bottom of the sawcut. The width shall be stamped on the heel part of the sawcut at the right side of the available space and placed so that bottoms of letters are within 1/16 inch of the top of plastic last. On three and four letter widths, the stamping may be done using 3A and 4A in lieu of AAA and AAAA. Characters for sawcut stamping shall be not less than 1/2 inch nor more than 3/4 inch in height and not less than 1/4 nor more than 3/8 inch in width. All sawcut stamping shall be readable when viewed from directly above the thimble. In addition to the stamping on the sawcut, the size and width designations shall be plainly rubber-stamped on the outside of the last, just ahead of the sawcut, 1-1/2 (± 1/2) inches from the top of the cone, to read horizontally. Characters for this stamping shall be the same size as for sawcut stamping. The width designation shall be

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on a line with the size designation. The size and width shall be clearly and legibly die stamped (incised) on the forepart of the last in the hollow area at the lowest point on profile line from cone to toe. Ink marking on plastic last shall be permanently set into the plastic by heat.

Marking for Identification. An identifying inscription shall be marked (incised) with a die stamp on the outside surface of each last, using not less than 3/8 nor more than 1/2 inch characters. The inscription shall be as follows: "W-F". In addition, the name or symbol of the supplier shall be plainly rubber-stamped on the inside surface of the back part of each last using the ink described above.

Models, Patterns and Templates.

Turning Models. Turning models for each of the sizes and widths required will be furnished by the Government. Turning models shall be in whole sizes in widths AAAA, AAA, AA, A, B, C, D, E, with the final measurements for the lasts of that size plainly marked on each model. Only two sizes shall be turned from any model, namely, the model size and one-half size up. No last shall be turned from a model of a different width. Models worn to the point of unserviceability in the production of properly graded lasts will be replaced by the Government. All models shall be returned to the Government at the close of the contract. Worn models that need replacement shall be marked "Unserviceable".

Last Bottom Patterns. Necessary turning bottom patterns and finished bottom patterns will be loaned to the supplier by the Government.

Templates. Templates for use in checking the lasts will be loaned by the Government to the supplier.

Measurements and Grading. The supplier shall check the turning models but shall make no modifications except for slight adjustments to fit the turning model bottom to the paper last bottom patterns furnished by the Government when this may be necessary due to swelling of the wood. The finished size 4 lasts shall conform to the measurements listed in table II. Size stick measurements on the size 4 last shall be taken from a point 1-3/4 inches up from the bottom of the heel curve; for size stick measurements on other sizes, the location of this point shall be raised 1/16 inch for each size larger and lowered 1/16 inch for each size smaller. Bottom of lasts shall grade 1/12 inch between widths of a given size. Lasts shall grade 1/3 inch between sizes in length. Variations in girth measurements for ball, waist, and instep shall not exceed 1/16 inch. Finished last bottoms shall conform to measurements of finished last bottom patterns furnished by the Government with tolerance of $\pm$ 1/32 inch in length and $\pm$ 1/48 inch in width.

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Table II. Table Of Measurements (Inches)

Size and width	Ball girth	Waist girth	Instep girth	Heel	Toe spring	Toe thickness	Stick length adjustment 1/
4AAAA	7-1/8	7-1/16	7-6/8	11-3/8	7/8	5/8	3-3/8 S
4AAA	7-3/8	7-2/8	7-7/8	11-4/8	7/8	5/8	3-3/8 S
4AA	7-4/8	7-3/8	8-1/8	11-5/8	7/8	6/8	3-3/8 S
4A	7-6/8	7-5/8	8-3/8	11-6/8	7/8	6/8	3-3/8 S
4B	8-1/8	7-7/8	8-5/8	11-7/8	7/8	6/8	3-4/8 S
4C	8-2/8	8-1/8	8-7/8	12	7/8	6/8	3-5/8 S
4D	8-4/8	8-3/8	9-1/8	12-1/8	7/8	7/8	3-6/8 S
4E	8-6/8	8-5/8	9-2/8	12-2/8	7/8	7/8	3-6/8 S

1/ For use and guidance of last manufacturer in grading lasts.

Workmanship. The plastic lasts shall be clean, free of plastic hairs, strings, flash or sprues and shall have no prominent turning gouges of any kind on the surface. All sharp edges in V-cut or sawcut shall be given a noticeable radius. Depressions, bumps, or holes of any kind arising from improper turning or interior voids in the block which appear on the surface of the last shall be cause for rejection.

Product Demonstration Model. When required, offerers shall submit samples of their intended production. The Product Demonstration Model will consist of units independently developed. Product Demonstration Models will be analyzed and evaluated by the government. Evaluation criteria and basis for award are contained in the solicitation. Product Demonstration Models will be retained by the government. One (1) Product Demonstration Model approved by DPSC under the resultant contract will serve as a manufacturing standard for the corresponding production items delivered under this contract. Offerers are advised that the Product Demonstration Model does not relieve the successful offerer of his responsibility to perform in accordance with the Commercial Item Description specified above.

Quality Assurance

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for, under the provisions of the contract. Reliance on contractor QA systems shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract.

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Examination. Last lots shall be inspected in accordance with "Sampling Procedures and Tables for Inspection by Attributes", ANSI/ASQC Z1.4, published by the American Society for Quality Control.

Visual Examination. The lasts shall be examined for the defects listed below. The lot size shall be expressed in units of lasts. The sample unit is one last.

Defects. The lasts shall be examined for the following defects: contains plastic hairs, strings, flash or sprues; depression, bump, hole or void; prominent turning gouge on surface; surface of last that will come in contact with the footwear not smoothly finished, i.e., protrusion, or rough area; part misplaced or out of alignment; heel end and forepart of last not joined flush, i.e., any step or ridge over 1/64 inch in the outside joint area that will come in contact with the footwear; part not a good flush fit where required, i.e., the breast of heelplate to plastic, and plug to heelplate; part loose, i.e., heelplate plug is not tight; surface of plastic plug in heelplate not even with surface of heelplate; part not recessed in plastic where required, or not recessed to the specified dimension; functioning part that will not operate as required; part missing or damaged; nail missing, head bent over, or protruding; nail misplaced; sharp edge; marking omitted, incorrect, illegible, incomplete, not accomplished in the specified manner, size of characters not as specified, or not in proper location.

Dimensional Examination. The finished lasts shall conform to the dimensions specified. Any dimension that is not within the established tolerance shall be classified as a defect. The sample unit shall be one last.

Packaging Examination. The fully packaged shipping containers shall be examined in accordance with defects listed below. The lot size shall be the number of shipping containers in the inspection lot.

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EXAMINE	DEFECT
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling
Content	Number of items container is more or less than required
	NSN shown on one or more items not as specified on shipping container
Palletization	Length, width, or height exceeds specified maximum requirements
	Pallet type not as specified
	Load not bonded as specified

Regulatory Requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

End Item Performance Testing.

Resistance to Impact. The finished last shall be tested by dropping an eight pound solid iron or steel ball so as to strike the last in the following spots (The ball shall be dropped from a height of two feet, measured between the bottom of the ball and the point of impact on the last):

a. With the last resting on its side, the ball shall strike over the rear hinge pin hole, over the front hinge pin hole, and at the edge of the ball line midway between the hinge and the toe.

b. With the last resting on the heel end, the ball shall strike the last on the tip of the toe.

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Tests shall be repeated three times in each spot. In the (a) position, repeat on both the outside and inside of the last. The last shall be inspected for breaking, splitting, cracking or other defects rendering the last unserviceable, other than directly at the impact area.

NOTE: The last may conveniently be held by a large C-clamp between the thimble and the heel plate and shall be placed on a concrete floor.

Hinge Performance. The last shall be placed on a regular last spindle and shall be broken and reclosed 100 times. This may be done by hand or by an appropriate hydraulic or mechanical mechanism. Examination should be made for defects developed by this test such as cracking, breaking, chipping or other defects which would make the last unserviceable.

Resistance to Tacks. The finished last shall be tested by driving 2-1/2 ounce insole tacks into the bottom surface of the last, 1/4 inch in from the edge, around the periphery from inside to outside ball area. The tacks shall be driven 1/4 inch deep and shall be spaced 1/2 inch apart. The finished plastic last shall show no splitting, chipping, cracking or other defects which would render the last unserviceable.

Preservation, Packing, Marking and Palletization. The preservation, packing, marking and palletization shall be as specified in the contract or order.

Source of Government Documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

Sources of Nongovernment Documents.

ANSI/ASQC Z1.4 - Sampling Procedures and Tables For Inspection By Attributes

(Applications for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

ASTM-D 1248 - Polyethylene Molding and Extrusion Materials

ASTM-D 3951 - Standard Practice for Commercial Packaging

(Applications for copies should be addressed to American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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	A-A-55172	93-2-4

3. DOCUMENT TITLE

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4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATIN: DPSC-FSSD (12-3-D)	(1) Commercial 215-737-8105	(2) AUTOVON 444-8105
c. ADDRESS (Include Zip Code) 2800 South 20th Street P. O. Box 8419 Philadelphia, Pennsylvania 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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